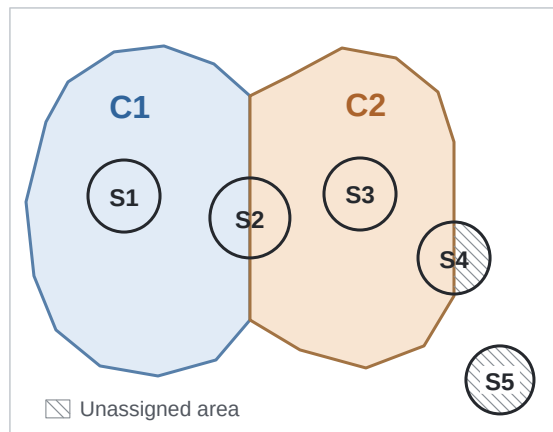


## a Registered masks + spot footprints



S2 shares  $x_2$ ; S4 retains unassigned area.

## b Cell-by-spot matrix A

$$a_{ci} = \text{area}(C_c \cap D_i) / \text{area}(D_i)$$

$D_i$ : spot footprint  $0 < \alpha, \beta < 1$

|         | S1 | S2         | S3 | S4        | S5 |
|---------|----|------------|----|-----------|----|
| C1      | 1  | $\alpha$   | 0  | 0         | 0  |
| C2      | 0  | $1-\alpha$ | 1  | $\beta$   | 0  |
| Outside | 0  | 0          | 0  | $1-\beta$ | 1  |

Outside area is not renormalized.

## c Feature aggregation

$$y_c = \sum_i a_{ci} x_i$$

$x_i = (x_{i1}, \dots, x_{iF})$ : feature counts;  $i$  = spot,  $f$  = feature,  $F$  unspecified.

$$C1 \quad y_1 = x_1 + \alpha x_2$$

$$C2 \quad y_2 = (1-\alpha)x_2 + x_3 + \beta x_4$$

Schematic footprints;  $\alpha$  and  $\beta$  are symbolic. S5 contributes to neither cell.